

Brandon Khumalo

BACKEND ENGINEER & DEVOPS ENGINEER

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PROFESSIONAL SUMMARY

Backend Engineer & DevOps Engineer with hands-on experience designing, building, deploying, and maintaining scalable APIs and production systems using Go, Python, Django, Django REST Framework, PostgreSQL, Docker, AWS, and CI/CD pipelines with GitHub Actions. Strong backend and infrastructure focus, with working frontend knowledge in React, Next.js, TypeScript, and JavaScript for cross-stack delivery. AWS Certified Cloud Practitioner and currently studying for the AWS Certified DevOps Engineer certification.

SKILLS

Programming Languages: Python, Go/Golang, JavaScript, TypeScript, SQL

Backend: Django, Django REST Framework, Echo, REST APIs, API design, authentication, authorization, background processing

Databases & Cache: PostgreSQL, MySQL, Redis, schema design, indexing, multi-tenant data models

Cloud & DevOps: AWS, Docker, Docker Compose, GitHub Actions, CI/CD, Nginx, Linux, SSL, deployment automation, production monitoring

Frontend: React, Next.js, Tailwind CSS, shadcn/ui, TanStack Query, React Context

Tools: Git, GitHub, VS Code, Swagger/OpenAPI, pytest, Postman

EXPERIENCE

Tishanyq Digital

Backend Engineer & DevOps Engineer | 2025 - Present

- Own AWS cloud infrastructure, including provisioning, configuring, and managing the services that run production applications.
- Design and maintain CI/CD pipelines with GitHub Actions and containerise services with Docker for repeatable, automated deployments.
- Build and deploy production-grade backend systems using Go, Python, Django, Django REST Framework, PostgreSQL, Redis, and REST APIs.
- Design scalable system architectures for client projects, internal products, SaaS platforms, dashboards, and API-driven applications.
- Optimise backend performance, database queries, application reliability, scalability, and resource usage across production systems.
- Implement secure authentication, authorization, role-based access control, API validation, rate limiting, and data protection practices.
- Configure and maintain production Linux servers, Nginx reverse proxies, SSL certificates, environment variables, and deployment scripts.
- Conduct code reviews, debug production issues, and implement solutions that balance speed, reliability, security, and maintainability.
- Work across backend, infrastructure, and frontend layers when needed using Go, Python, Django, JavaScript, React, TypeScript, and PostgreSQL.
- Collaborate with stakeholders to translate business requirements into reliable technical systems, production workflows, and maintainable software.
- Ensure application security, data integrity, and alignment with regulatory standards, client requirements, and industry best practices.

SELECTED PROJECTS

My School Hub - Multi-tenant SaaS School Management Platform

Tishanyq Digital | 2025 - Present | myschoolhub.co.zw

Multi-tenant SaaS school management platform serving academics, boarding, clinic, finances, HR, library, parent communication, payment workflows, and reporting across 13 user roles. Built with a Go and Django backend architecture, AWS infrastructure, and production-focused deployment workflows.

Backend Engineering

- Designed and built a polyglot backend architecture using a Go API gateway in front of a Django REST Framework core, with specialised Go services for integrations and bulk processing.
- Built a Django backend with 45+ domain models covering academics, students, staff, finance, boarding, clinic, library, reports, permissions, and school-level configuration.
- Implemented a Go API gateway to centralise authentication checks, request routing, rate limiting, CORS handling, audit logging, and communication with downstream services.
- Designed a row-level multi-tenant data model where school-specific data is isolated using school-scoped models, custom managers, and role-based access rules.
- Built a role-based permission system for 13 user roles with per-page read/write permissions, backend enforcement, and dashboard-level access control.
- Developed a streaming bulk-import service for students, fees, and academic results with batch inserts, conflict handling, and duplicate prevention.

- Built integration workflows for PayNow payments, Resend transactional email, WhatsApp Business API communication, and PDF report-card generation.
- Designed a configurable report-card engine supporting school branding, assessment plans, weighted marks, teacher feedback, approval workflows, and released PDF reports.
- Built boarding operations features including dormitory management, bed assignments, roll call, lights-out tracking, exeat workflows, tuck-wallet transactions, medication schedules, dorm inspections, and wellness check-ins.
- Implemented audit logging for key platform actions, including create, update, delete, login, suspend, and approval events, with changed-field tracking and request metadata.

Infrastructure / DevOps

- Designed and deployed the production infrastructure on AWS for a multi-service SaaS platform with separate backend, gateway, worker, database, cache, media, and frontend concerns.
- Built a production Docker Compose stack running the Go gateway, Django backend, Go services, background workers, scheduled jobs, and Nginx reverse proxy.
- Configured Nginx as the production entry point with reverse proxy routing, SSL termination, domain configuration, and secure request forwarding.
- Implemented SSL certificate setup and auto-renewal using Let's Encrypt and Certbot, including reload hooks and boot-resilient configuration.
- Built GitHub Actions CI/CD pipelines to test, build, package, and deploy backend services to production.
- Wrote deployment scripts supporting backend deployment, full deployment, service restarts, environment updates, and rollback workflows.
- Designed a scaling path from a single-server production setup to a container-orchestrated architecture with load balancing, rolling deployments, and autoscaling.
- Documented deployment procedures, troubleshooting steps, scaling phases, cost considerations, SSL recovery, and infrastructure migration plans.
- Applied secure infrastructure practices including least-privilege access, private service communication, environment variable management, and security-group based access controls.
- Improved deployment reliability by separating application services, background tasks, reverse proxy configuration, media handling, and database responsibilities.

Tech Stack: Python, Django, Django REST Framework, Go, Echo, PostgreSQL, Redis, Docker, Docker Compose, GitHub Actions, AWS, Nginx, Let's Encrypt, Certbot, PayNow, Resend, WhatsApp Business API

Laptop World - Full-Stack E-Commerce Platform

Client Project | March 2025 - April 2025 | laptopworld.co.zw

Production e-commerce platform for a Zimbabwean technology retailer using a decoupled Next.js frontend and Django REST Framework backend.

- Built and deployed a production e-commerce platform end-to-end using Next.js App Router, Django REST Framework, PostgreSQL, Docker, and a decoupled frontend/backend architecture.
- Implemented session-based cart synchronisation using UUID session keys in a custom request header, allowing customers to shop without creating an account.
- Integrated PayNow Zimbabwe payment processing with asynchronous payment confirmation after order creation.
- Engineered automatic image compression on Category, Product, and ProductImage model saves to improve media performance at the data layer.
- Designed a 7-stage order lifecycle covering awaiting payment, paid, processing, out for delivery, delivered, collected, and cancelled states.
- Integrated Resend transactional email workflows for order confirmation and customer communication.
- Built a Google Maps delivery address picker with GPS coordinates stored on the order model and automatic delivery fee calculation based on location.
- Designed backend models, serializers, API endpoints, validation logic, and admin workflows to support real-world e-commerce operations.

Tech Stack: Next.js, TypeScript, Django REST Framework, PostgreSQL, TanStack Query, React Context, Zod, react-hook-form, Tailwind CSS, shadcn/ui, Framer Motion, Docker, PayNow, Google Maps API, Resend

TumaGo - Full-Stack Mobile Package Delivery App

Personal Project | April 2025 - June 2025

Mobile logistics platform with booking flows, driver matching, GPS tracking, delivery status updates, and push notifications.

- Architected a hybrid backend using a Go API gateway, Go location and matching services, and Django REST Framework for business logic, admin, and payment workflows.
- Built live driver tracking using location updates, Redis GEO indexing, and PostgreSQL persistence for real-time delivery visibility.
- Implemented a driver matching engine using asynchronous workers and Redis GEO proximity search to surface the nearest available driver by vehicle type.

- Developed native Android client and driver applications in Java with booking flows, live GPS tracking, push notifications, earnings dashboards, and Google Maps/Places integration.
- Implemented secure authentication, per-IP rate limiting, protected internal endpoints, graceful error fallbacks, and notification workflows.
- Built a notification service and integrated Firebase Cloud Messaging for delivery status updates, driver messages, and customer notifications.
- Containerised backend services with Docker and prepared deployment workflows for production environments.
- Added API documentation and development tooling to support maintainability, testing, and future scaling.

Backend: Go, Echo, Python, Django REST Framework, Redis, MySQL, Firebase Admin, AWS

Frontend: Java Android, Retrofit, OkHttp, Firebase Messaging, Google Maps SDK, Places API

DevOps & Tools: Docker, GitHub Actions, Prometheus, Grafana, Swagger/OpenAPI

EDUCATION

ALX Africa

Certificate in Pro Backend Development | Jan 2025 - Aug 2025

CERTIFICATIONS

- AWS Certified DevOps Engineer - Amazon Web Services, In Progress
- AWS Certified Cloud Practitioner - Amazon Web Services, June 2024
- Certificate in Pro Backend Development - ALX Africa, Aug 2025